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Influence of deficit irrigation and biochar amendment on growth, physiology, and yield of cucumber in West Texas

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Despite growing interest in adopting water-conserving strategies such as deficit irrigation (DI) and biochar amendment in arid and semi-arid regions, there are remarkably few studies conducted on the integration of these two strategies in water limited West Texas region. The aim of this study was to assess the interactive effects of DI and biochar amendment on saturated hydraulic conductivity of soil and the growth, physiology and yield of cucumber (*Cucumis sativus*) in West Texas. A split-plot design was used to randomize four irrigation treatments (with two levels before and after mid-season) as the main plot [I1: 100% crop evapotranspiration (ETc)-100% ETc, I2: 80% ETc-60% ETc, I3: 60% ETc-80% ETc, I4: 40% ETc-40% ETc] and three biochar rates as subplot [0 t/ha, 15 t/ha, and 20 t/ha] with four replications. Irrigation level I4 negatively impacted chlorophyll content, stomatal conductance, transpiration, photosynthesis, and intrinsic water use efficiency (iWUE) when the weather was hot and dry in 2022 compared to other treatments. Leaf area index (LAI) under I4 declined significantly by 52% and the final dry biomass by 47% compared to I1 across two years, although I3 reduced biomass by 17% only in 2022. As compared to I1, the yield penalties due to water stress in I2, I3, and I4 were 14, 16, and 34%, respectively, across two years. Although biochar rates 15 t/ha and 20 t/ha significantly increased the saturated hydraulic conductivity of soil by 1.6 and 2.2 folds in 2022 compared to the unamended control but biochar's impact on cucumber growth, physiology, and yield was minimum during both years. Hence, our findings suggest that I2 can be adopted as an alternative to full irrigation which requires 25% less water and has 14% narrow yield gap. Biochar studies are needed to assess whether added biochar has an apparent long-term effect on improving cucumber yield.

Keywords Ogallala aquifer, Irrigation, Biochar, Fruit yield, *Cucumis sativus*

The Ogallala aquifer supplies water to eight states[1]in the U.S. to meet water needs for agriculture and public use[2]. The Texas High Plains (THP) is a semi-arid region that satisfies 90% of its irrigation needs from the Ogallala aquifer[3]. However, the water withdrawal rate of the aquifer exceeds its recharge rate, creating a depletion of the water table in the region, which makes irrigated agriculture difficult to sustain[2]. Therefore, there is a need to develop water-conserving management strategies to sustain crop production and the longevity of the Ogallala aquifer in the water-scarce THP[4].

Deficit irrigation (DI) is a widely practiced strategy in water-limited agriculture regions to manage and conserve water resources. Under DI practices, the crop is supplied with less water than its ET water demand that can enhance water use efficiency (WUE) calculated as yield per amount of water used[5]. Deficit irrigation essentially helps to save a considerable amount of water in irrigated agriculture especially where water management and conservation practices are highly needed[6]. Deficit irrigation is highly recommended in semi-arid and arid regions where the main goal is to enhance yield per unit of water (i.e., water productivity) rather than yield per unit of area or land (i.e., land productivity)[7]. The

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response of DI depends on the crop, crop stages, as well as varieties[8]. Kirda[6]reported that DI could stress the crop by limiting the water supply during a certain growth period or the whole crop season. The crop requires different amounts of water at various growth stages, i.e., vegetative and reproductive stages[9], and hence, the growth stage-based DI management has been shown to be successful in many crops[10,11]. When knowledge of irrigation scheduling is limited, the inappropriate amount of water supplied to the crop may not be optimally utilized by the crops, resulting in low water productivity. However, before applying any irrigation strategies, it is imperative to understand the physiological response of crops to water stress at different growth periods[12].

Water stress leads to many biochemical and physiological changes in plants at different growth stages, ultimately bringing morphological changes[13]. Different crops respond differently to water stress based on their water stress tolerance ability. Generally, when the plants are stressed, the stomata tend to close, which limits the gaseous exchange, decreases CO_2assimilation, and reduces transpiration. This decreases leaf-level photosynthesis, and ultimately, overall growth is reduced[14]. These parameters may alter depending on the growth stage of the crop under water stress. Prolonged water stress during the critical growth period of the crop can be detrimental; hence, recognizing that sensitive growth stage is imperative while adopting DI. Any changes in physiological parameters during the critical growth period can be a good indicator for estimating the productivity of crops under water-limited conditions. This is because water stress in this stage alters many morphological and physiological parameters that significantly determine yield[15].

Cucumber (*Cucumis sativus*) is a shallow-rooted moderate water stress tolerant summer crop and has its maximum rooting at 20–30 cm depth with negligible root fiber extension below 60 cm[16,17]. It is grown on ~2500 ha in Texas, where west Texas alone shares 50% of the cultivated area[18]. According to the USDA Vegetable Summary report of 2022, Texas alone shares more than 9 million US dollars in value for the national economy from the fresh cucumber market[19]. However, the production of cucumber has been impacted by extreme climatic conditions and water scarcity in the West Texas region[4]. It has been reported that significant yield reduction occurs with prolonged water deficiency during cucumber production[20]. However, Mao, et al.[17]found that WUE was higher with greater irrigation water amount supplied from crop establishment to first fruit ripening but reduced afterward until the end of the fruit harvest. It has been described that cucumbers do not need continuous high irrigation throughout their growth stage. Rahil and Qanadillo[21] reported that 70% ET replacement produced a 2.2 t/ha higher yield than 100% ET replacement irrigation in cucumber. In a greenhouse experiment with Japanese cucumber, Oliveira, et al.[22]found that stressing the crop at the reproductive stage reduced yield significantly compared to the vegetative stage. During flowering and fruiting, more energy exerted by a plant was directed to absorb water from the soil, which caused the flower abortion and small fruit sizes. A greenhouse study was conducted at 40, 60, 80, and 100% ET replacement irrigation levels at different growth stages of cucumber, i.e., initial, developmental, middle, and late, and found that initial and late stages were the most sensitive, the developmental stage being the highest tolerant stage for DI in terms of yield[23]. A cucumber experiment conducted by Parkash, et al.[4] showed that cucumber can have a similar yield at 80% ET compared to 100% ETc. This finding has provided more room to develop a growth stage-based ET requirement rather than a seasonal ET strategy to save more water in this region.

Other water-conserving farming strategies, such as soil amendments, could be integrated with DI for a water-limited or water-stressed production system[24]. Soil amendments are farm inputs applied to soil to improve soil properties, which could later help crop production. Biochar, a promising soil amendment, is a highly porous stable carbon compound obtained from burning plant residues in a limited oxygen environment at high temperatures[25]. As biochar has a greater capacity to retain water in the soil, its potential for adoption as a soil amendment is increasing in water-scarce regions[26,27]. It has been reported that in 90% of studies, the application of biochar has increased water holding capacity in different soil types[28]. Although the application of biochar has resulted in an enhanced favorable response in soil, there is no consensus as to the upper or maximum limit of biochar application rates or amounts, especially depending on the variability of soil and crop type[29]. Moreover, the source of biochar (feedstock), pH, and nutrient content of biochar also determine the application rates of biochar[30]. Liu, et al.[31]mentioned that an application rate below 30 t/ha can have positive effects on crop productivity but largely depends upon the crop type. Lehmann, Gaunt and Rondon[32]reported that 55 t/ha of biochar is the upper limit for any crop. However, Filiberto and Gaunt[29]recommended 25 t/ha as the upper limit for crop production. Since many studies investigating different application rates of biochar as soil amendments have reported varied and contrasting results on the upper or maximum limits of biochar application rates[33,34], it is difficult to generalize recommendations for various crops based on individual studies. Hence, there is a need for a site-specific recommendation rate of biochar for growers that could help increase crop productivity.

Among different soil hydraulic parameters, saturated hydraulic conductivity (Ksat) is of immense importance in determining soil quality associated with water flow in the soil[35]. In a review, Blanco-Canqui[28]highlighted that biochar reduced Ksat in 12 coarse-textured soil studies by 7–2270% but increased Ksat in 8 fine-textured soil studies by 25–328% because of the high coefficient of variation in soil properties. A greenhouse study in sandy and sandy loam soils with rice husk biochar at a rate of 0, 0.1, 0.5 and 1% (w/w) showed an increase in Ksat value by 54–78% in sandy soil and 82–148% in sandy loam soil with >0.5% (corresponding to 10.5 t/ha) biochar rates due to enhanced macro-aggregation[36]. Another study by Toková, et al.[37] revealed that biochar application at 20 t/ha had the greatest improvement in soil Ksat by 464 and 350% under control (i.e., without biochar application or non-amended soil) and 108 kg/

ha fertilizer application, respectively, in silty loam soil. These reports show that biochar response can vary depending on the soil texture and application rate. Therefore, while biochar is broadly purported to provide agricultural benefits via biochar-induced changes in soil hydraulic properties in many studies, a site-specific biochar application recommendation is still needed in the West Texas region, especially after examining its impact on soil hydraulic properties, such as the Ksat of soil, under semi-arid conditions of West Texas.

Many pieces of literature repeatedly examined the effect of either biochar amendment or DI strategy individually on different crops. However, Singh, et al.[27] emphasized that field studies integrating these two management practices are still scarce, especially in semi-arid regions. To the best of our knowledge, this study is the first instance that has implemented the combination of these two well-recognized water-saving strategies to grow cucumbers under semi-arid conditions in West Texas. This novel research will help to explore the feasibility of irrigation and soil amendment in sustainable production of cucumber which is rarely investigated area in the region. Our hypothesis for this study is that the integration of DI and biochar would significantly enhance cucumber productivity. Therefore, this study aimed to (i) investigate the effect of biochar rates on the Ksat of soil, and (ii) evaluate the integrated effects of DI and biochar rates on growth, physiology, and yield of cucumber in West Texas.

Materials and methods

Study location

A two-year field study was conducted from April through August in 2021 and 2022 at Quaker Research Farm of Texas Tech University, Lubbock, TX (33° 36′ 18″ N, −101° 54′ 26″W, and 992 m above sea level). The study site was a semi-arid region with an average annual evapotranspiration of 1,501 mm and an average annual rainfall of only 469 mm[38]. The study location had an average annual low and high temperatures of 8.3℃ and 23.3℃, respectively. The soil type was deep, well-drained, friable, dark grayish brown, and neutral to moderately alkaline with a slope gradient of 0–1%[39]. The soil texture (0–10 cm), measured on collected soil samples at the beginning of the experiment in 2021 using the hydrometer method[40,41], was sandy clay loam with a particle size distribution of 55% sand, 11% silt, and 33% clay. The description of soil nutrient status is presented in Table 1.

Experimental design and cultivation practices

The experimental units were randomized four times in a blocked split-plot design with four growth-stage based irrigation levels: I1 [100% crop evapotranspiration (100% ETc) replacement from crop establishment to mid-season (50–55 days after planting) CMS-100% ETc replacement from mid-season to maturity (MSM)], I2 (80% ETc CMS-60% ETc MSM), I3 (60% ETc CMS-80% ETc MSM), I4 (40% ETc CMS-40% ETc MSM) as main plot factor and three biochar rates (0 t/ha,15 t/ha, and 20 t/ha) as subplot factor. Each plot was 4 m wide and 6 m long with raised beds (i.e., ridges) of approximately 50 cm wide and 15 cm high. The irrigation was applied via the sub-surface drip irrigation system. The drip tapes were buried in the soil at a depth of approximately 30–35 cm from the top of the ridge. The experimental field consisted of four irrigation zones, each equipped with an independent flow control system or regulator and a flowmeter, to apply irrigation treatments (i.e., I1, I2, I3, and I4). Within an irrigation zone, each plot was separated by 1.5 m wide alleys.

At the start of the experiment each year (January-February of 2021 and 2022), the experiment field was plowed with a tractor-mounted disc plow during winter to break large clods and obtain fine tilth. Pre-plant fertilizer URAN 32 (NPK 32-0-0, Nutrien Ag Solution, Loveland, CO) was applied through fertigation once before planting based on a soil fertility test at the rate of 80 kg N/ha. Hardwood biochar prepared from Oak tree (Wakefield™ BioChar, Valdosta, GA) was used for the study, and the physical and chemical properties of this hardwood biochar can be found in Singh, et al.[13]. The biochar was spread evenly on the soil surface in treatment plots and incorporated to a depth of approximately 15–20 cm, along with the preparation of raised beds using a tractor-mounted rotary tiller on March 18, 2021. The cucumber cultivar Bristol (Rupp Seed Inc., Wauseon, OH) was selected for this study due to its specific adaptation to the semi-arid environments of West Texas. The

Characteristics of soil	Parameters value
pH	8.3
Cation Exchange Capacity (CEC)	17.8 cmol (+)/kg
Organic Matter	0.20%
NO_3^-Nitrogen (N)	8 mg/kg
Phosphorus (P)	35 mg/kg
Potassium (K)	503 mg/kg
Magnesium (Mg)	646 mg/kg
Calcium (Ca)	2236 mg/kg

Table 1. Soil chemical properties (0–10 cm) at quaker research farm, Lubbock, TX. Note: Procedure for analysis of soil pH (Saturated paste), CEC (Ammonium replacement), organic matter (Loss of ignition at 360℃), NO_3^- Nitrogen (2.0 N KCl/Cadmium Reduction), and phosphorus, potassium, magnesium, calcium (1.0 N ammonium acetate @ pH 7.0) reference to North American Proficiency Testing (NAPT) Program (https://www.naptprogram.org/).

planting was done on June 10, 2021, and June 6, 2022, by a tractor-mounted four-row planter with row-to-row spacing of 1 m at the rate of 45,000 seeds/ha.

Cucumber, grown under each of the four DI treatments (I1, I2, I3 and I4), was irrigated based on the ETc calculated by multiplying reference evapotranspiration (ETo) by crop coefficient (Kc). The daily ETo was calculated from weather data using the Penman-Monteith method[42]. The daily weather data was collected from a weather station (Davis Wireless Vantage Pro2, Hayward, CA) installed near the study location. The growth stage-specific crop coefficient of cucumber was $K_{c\ initial}$ = 0.45 (0–20 Days After Planting; DAP), $K_{c\ crop\ dev}$, = 0.70 (21–50 DAP), $K_{c\ mid}$ = 0.90 (51–90 DAP), $K_{c\ late}$= 0.75 (91–105 DAP)[4,43]. Each irrigation zone had a separate irrigation valve which emitted approximately 1 inch of water in 24 h. Based on the water requirement, irrigation valves were opened for a certain amount of time to supply the required volume of water at every irrigation event. Irrigation was applied and adjusted based on the calculated ETc requirements and rainfall events from the previous week.

Measurements and data collection

Soil sampling for saturated hydraulic conductivity (Ksat)

To assess the effect of biochar on soil hydraulic property, soil core samples from 0 to 5 cm soil depth were collected in a stainless-steel soil core ring of 5 cm in height ×8 cm in diameter. The soil core ring was first vertically pressed into the soil by hand and then placed a block of wood on top and pushed further by steady hammering till the core sampler was driven down to the surface of the soil. Both ends of the soil core ring were closed by plastic end caps to help prevent breakage of the soil core and preserve soil sample integrity. Soil core samples were collected from eight replicated plots under each biochar treatment. After leveling the soil at each end of the soil core ring, one end of the ring was placed over a gasket with the porous plate, and it was saturated in deionized water for 24 h. The core was placed over the measuring dome with a crown and screw cap over the core in the Ksat benchtop instrument (Meter Group, Inc., Pullman, WA, USA). Using the Falling Head method, water from the burette was allowed to pass through the saturated soil in the sample core until the steady-state flow (cm/day) was not achieved and the pressure head difference reached a stop criterion[44].

The Ksat works based on the Darcys' equation under the principle of hydraulic gradient which is explained by the following Eqs. 1 and 2 adapted from the[44].

$$\mathrm{Ksat} = -\frac{(\mathrm{L.V})}{(\mathrm{H.A.T})} \tag{1}$$

$$\mathrm{Ksat} = 2.303(\mathrm{aL/At})\log_{10}(\mathrm{Ho/Ht}) \tag{2}$$

Where, L = length of the sample, V = volume of water passing through the core soil sample, H = height of the water column, A = cross-sectional area of the core soil sample, T = time to travel water through the soil sample, a = burette cross-sectional area, L = length of soil sample, Ho and Ht = initial and final hydraulic head difference t = time taken to drop pressure head from Ho to Ht in burette.

Plant physiological parameters

To assess the effect of DI and biochar rates on the physiological responses of cucumber, a portable photosynthesis system (Model LI-COR-6800, LI-COR Biosciences, Lincoln, NE) was used to measure net photosynthesis (Pn), stomatal conductance (gs), transpiration (E) and instantaneous water use efficiency (iWUE). All these leaf-level measurements were conducted on the two youngest, fully expanded, sunlit leaves for two random plants in each treatment plot. The photosynthesis system was set at a steady state at 1,500 μmol m^{-2} s^{-1} photosynthetic active radiation (PAR), 400 μmol mol^{-1} reference CO_2 concentration, 700 μmol^{-1}air flow rate, 65% of relative humidity while switching the temperature control off[13]. The measurements of plant physiological parameters were taken between 10:00 am and 2:00 pm local time when the leaves were fully active, and PAR was high at 33, 62, and 72 DAP in 2021, and 42, 71, and 92 DAP in 2022, respectively. The leaf chlorophyll content (μmol/m^2) was also measured using the portable MC-100 chlorophyll concentration meter (Apogee Instrument, Logan, UT) on the same observation dates (i.e. 33, 62, and 72 DAP in 2021, and 42, 71, and 92 DAP in 2022) in both years.

Leaf area index, biomass, and yield

The LAI was measured in each treatment plot using the AccuPAR Ceptometer (LP-80, Decagon Device Inc., Pullman, WA) at local noon time when the sun was approximately perpendicular to the cucumber plants in rows. The measurements of LAI were made at 33, 62, and 72 DAP in 2021, and 42, 71, and 92 DAP in 2022, respectively. The plants (leaves and vines) were oven-dried at 70℃ for 72 h to achieve a constant weight. The plant dry biomass was then recorded using a weighing scale at 72 DAP and 92 DAP in 2021 and 2022, respectively. Matured fruits of marketable size were harvested once a week after 44 DAP in 2021 and 48 DAP in 2022 from an area of 6 m^2 in each plot, and the total number of fruits (fruits/ha) and the total fruit yield (t/ha) were recorded. The average weight per individual fruit was calculated by dividing the total fruit weight by the total number of fruit weighed in each plot.

Statistical analysis

The collected data were analyzed using the analysis of variance (ANOVA) with a split-plot design in R-Studio version 3.5.2[45]using the Agricolae package version 1.2–8[46]. Data for each measurement date (i.e., DAP) and each year were analyzed separately. The treatment means were separated using the Fisher's protected least significant difference (LSD) test at a 0.05 level of significance. There were no interactions between irrigation and biochar

factors for the data; hence, only the main effects are presented and discussed. The graphs were prepared using SigmaPlot software version 14 (Systat Software, San Jose, CA).

Results and discussion

Weather conditions and irrigation

The average temperature and relative humidity (RH) were 26.3℃ and 58.8% in 2021 (Fig. 1), and 27.1℃ and 49.7% in 2022 (Fig. 2), respectively. The daily average solar radiation received was 22.4 MJ/m^2 in 2021 and 23.1 MJ/m^{2}in 2022. The total rainfall amounts during the study were 149.8 and 118.7 mm in 2021 and 2022,

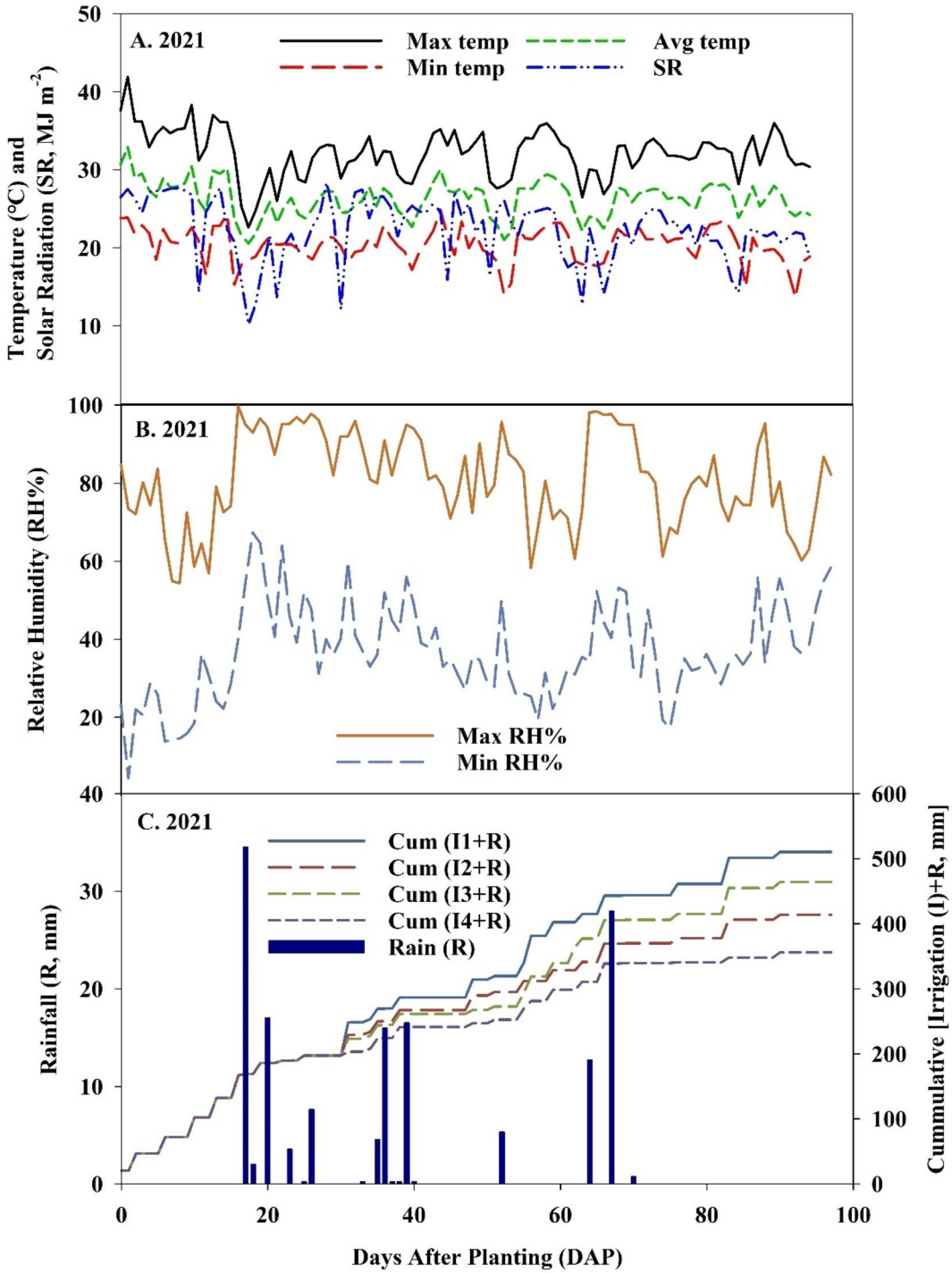

Fig. 1. Weather conditions (**A**) daily maximum, minimum, average temperature and solar radiation (SR), (**B**) daily maximum and minimum relative humidity (RH), and (**C**) rainfall and cumulative (rainfall + irrigation) during the 2021 growing season in Lubbock, TX.

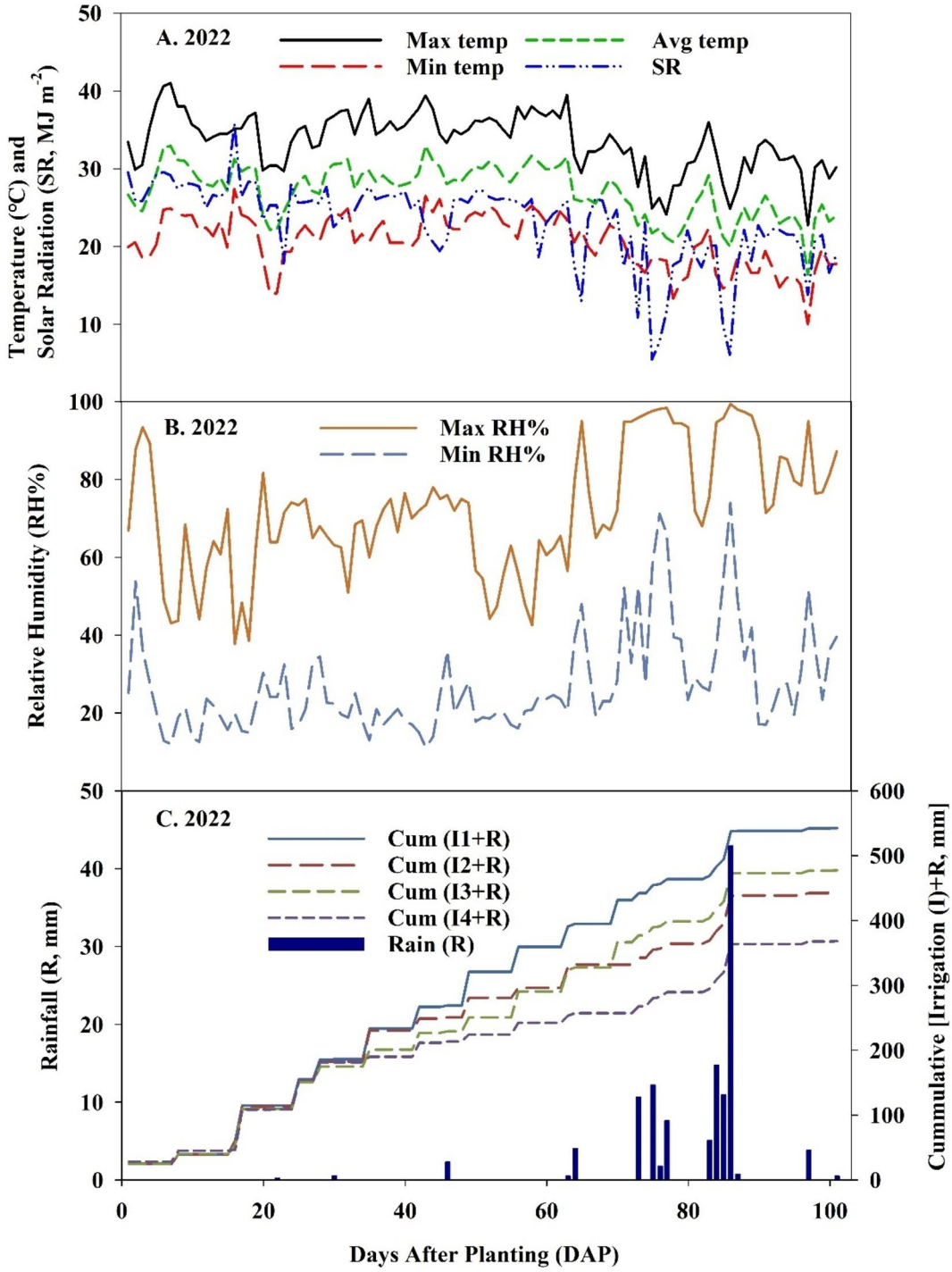

Fig. 2. Weather conditions (**A**) daily maximum, minimum, average temperature and solar radiation (SR), (**B**) daily maximum and minimum relative humidity (RH), and (**C**) rainfall and cumulative (rainfall + irrigation) during the 2022 growing season in Lubbock, TX.

respectively. The data showed that the second year (2022) was comparatively hotter and drier than the first year (2021). In 2021, rainfall was mostly received in the early and mid-stage, while it was majorly concentrated during the later growth stage of the cucumber plants in 2022. Relative to 2021, the uneven rainfall distribution in 2022 exposed the cucumber plants to more stressed conditions during their early-mid growth stages. Similarly, plants started to decay at a later stage in 2022, and the high rainfall events caused leaves to die and fall, resulting in fewer harvest events (only six times) compared to 2021 harvests (seven times)[47]. The total reference evapotranspiration (ETo), which was used in calculating ETc, over the growing season was 702 mm in 2021, while it was 737 mm in 2022. This resulted in variations in the amount of irrigation applied in both years for different DI treatments for cucumber production[4]. In 2021, a total of 341 (I1), 249 (I2), 306 (I3), and 185 (I4) mm of irrigation amounts

were applied. The total amounts of irrigation applied to different treatments in 2022 were higher, i.e., 452 (I1), 353 (I2), 374 (I3), and 283 (I4) mm, owing to the lower amount of rainfall and higher ETc requirement of the cucumber plant. The different DI treatments evaluated in this study resulted in irrigation water saving of 27% (I2), 10% (I3), and 46% (I4) in 2021, and 22% (I2), 17% (I3), and 37% (I4) in 2022, as compared to the control (I1).

Effect of biochar amendment on saturated hydraulic conductivity of the soil

There was no significant difference among biochar treatments for Ksat in 2021, but it significantly improved in biochar-amended treatments (15 and 20 t/ha) in 2022 (Fig. 3). The biochar treatments 15 t/ha and 20 t/ha increased Ksat by 1.6 and 2.2 folds, respectively, compared to 0 t/ha (i.e., the control) in 2022. The Ksat of the soil, which relates soil water flow rate to the hydraulic gradient, is a measure of the ease of water movement through the soil and considered one of the metrics of soil physical quality[35,48]. Zhou, et al.[49]reported that the porous compound biochar reduced the bulk density of soil, improved soil porosity, and ultimately increased the hydraulic property. The porous biochar that was added in our study might have contributed the porosity of the soil enhancing the pathway for water to flow in a soil. Blanco-Canqui[28]reviewed different studies that presented varying effects of biochar amendments on soil hydrological properties and reported that biochar influenced Ksat in coarse-textured soils to a greater degree than fine-textured soils. As mentioned in earlier, soils at the study site were characterized by 33.3% clay content and the added porous compound biochar most likely enriched the soil with the macro- and mesopores, thereby enhancing the hydraulic property of the soil by which it permits the movement of water through it (i.e., hydraulic conductivity). Our finding is consistent with a previous study, in which the wood biochar at 16 t/ha increased the Ksat by 176% compared to 0 t/ha (non-amended) in clay loam soil[50]. The effect of biochar in our study increased the Ksat by 61% and 121% in 15t/ha and 20 t/ha biochar in 2022, respectively in sandy clay loam soil. This difference in the enhancement due to biochar could be caused by different soil types. Another study by Lim, et al.[51] also reported an increase in Ksat under 1% and 2% (w/w) biochar amendment in clay loam soil compared to coarse sandy soil, underscoring the effect of soil textures on biochar-induced changes in soil Ksat. Notably, while most of the previous studies are limited to laboratory or greenhouse or controlled conditions, the findings in this study further clarified our understanding of the changes in soil Ksat associated with biochar addition under semi-arid field conditions. The insignificant difference in soil Ksat among treatments in the first year could be explained by the fact biochar might take more time to reach its full potential since the soil samples for determining Ksat were collected after five months of biochar incorporation. There is still no consensus on exactly how long it takes for the added biochar to be integrated with the soil to bring the most benefits to it, and the literature provides contrasting results in different studies. Wang, et al.[52] reported no significant alternation in Ksat in expansive clayey soils after one year of biochar application, whereas Jačka, et al.[53] reported significant benefits due to biochar on Ksat within some hours of application in sandy soil loam. Ouyang, et al.[54] mentioned that the addition of biochar significantly improved the macro

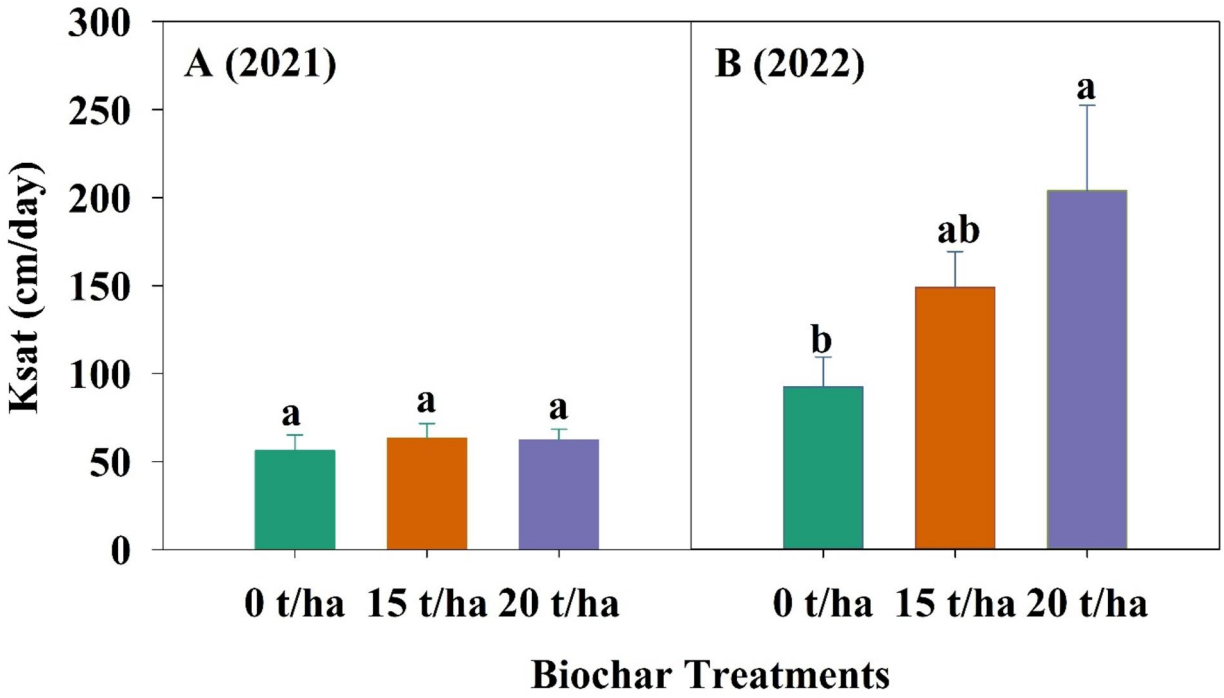

Fig. 3. Effect of biochar on saturated hydraulic conductivity (Ksat) of soil in 2021 (**A**) and 2022 (**B**) in Lubbock, TX. The lines on top of the bars indicate standard errors and different alphabets over the bars indicate significant difference at $P \leq 0.05$ level of significance.

aggregates and increased the saturated hydraulic conductivity of the soil once biochar settled in the soil. So, depending upon settling time and soil types, the effects of biochar on Ksat vary.

Responses on physiological parameters under integrated strategies of deficit irrigation and biochar soil amendment

Chlorophyll content

Studies have indicated that water stress can bring varying effects on leaf chlorophyll content because it can either increase[55], decrease[56], or have no effect[57] for various crops. In our study, we found that moderate plant water stress induced by DI levels, i.e., especially I2 and I3 treatments, can occasionally increase the leaf chlorophyll content in cucumber compared to non-stressed (the control I1) and highly stressed (I4) plants in 2021 and 2022 (Fig. 4A and B). The chlorophyll content in DI treatments (I2, and I4) was found to be significantly higher compared to I1 at 33 DAP but remained unaffected for the rest of the growing period in 2021 (Fig. 4A). It could be due to the occurrence of frequent rainfall events during the 2021 growing season, which might have resulted in relatively lower water stress conditions and contributed to relatively similar leaf greenness. With a given rainfall event that could favor the optimal soil water status in the root zone, plant uptakes of nitrogen (N) and magnesium (Mg) available in the soil could contribute to leaf sizes and help optimize chlorophyll synthesis[58]. However, in 2022, the leaf chlorophyll content was found to be enhanced, especially in DI treatment I3, but reduced to the least in I4 when compared with I1 at 70 DAP. At 70 DAP and 90 DAP in 2022, the significant drop in chlorophyll content in I4 was likely associated with lower N uptake rates[59,60] as there might be less availability of water to solubilize N and transport it to the plant. Our results agree with the findings of a previous study, in which the leaf chlorophyll content in sweet corn (*Zea mays*L.) was significantly dropped at higher water

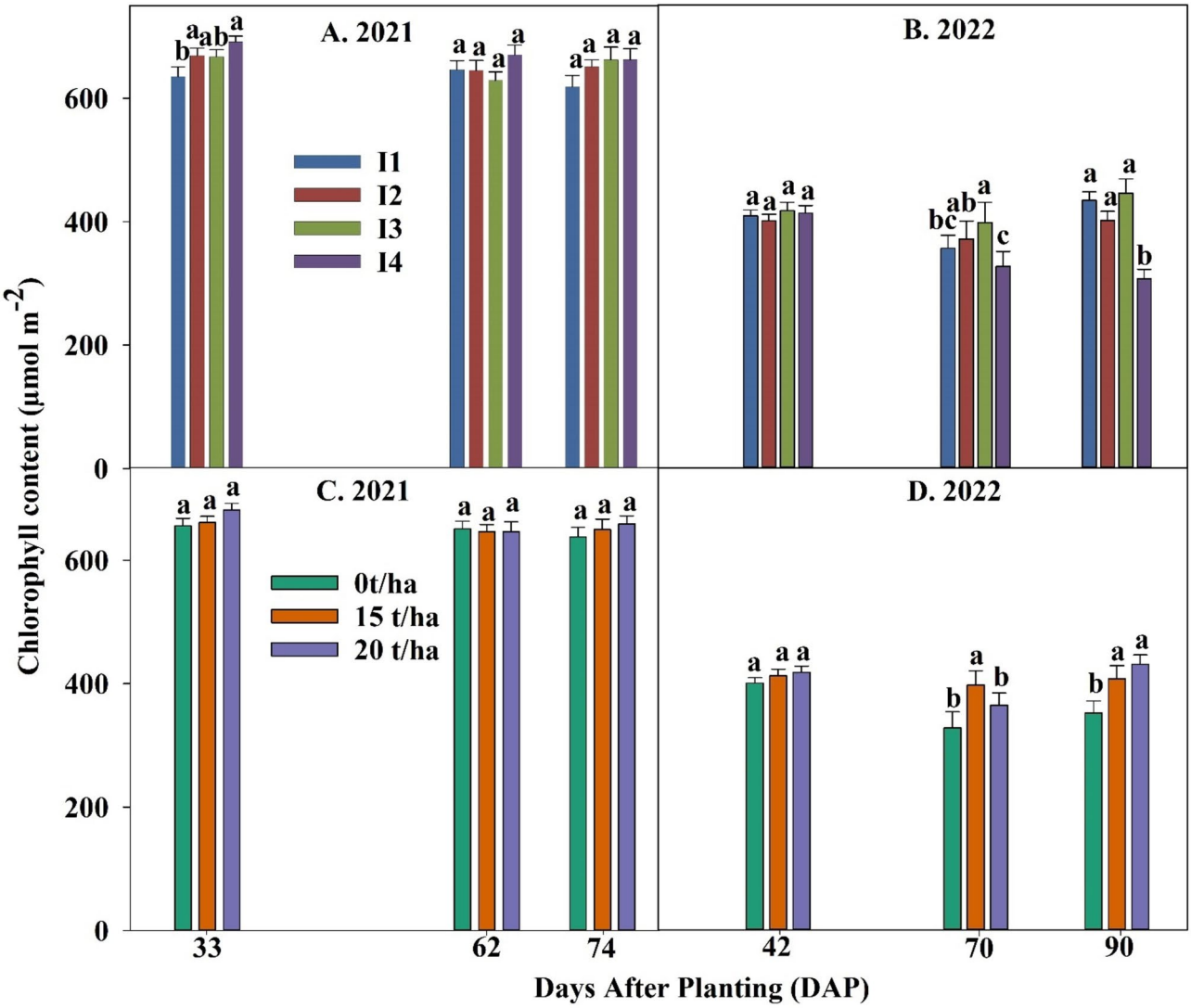

Fig. 4. Effect of deficit irrigation and biochar rates on chlorophyll content of cucumber in 2021 (**A** & **C**) and 2022 (**B** & **D**) in Lubbock, TX. The lines on top of the bars indicate standard errors and different alphabets over the bars indicate significant difference at $P \leq 0.05$ level of significance.

stress (40% ETc throughout the crop season) compared to the control (100% ETc)[13]. The decline in chlorophyll content in I1 at 33 DAP in 2021 and 70 DAP in 2022 could be due to excess water in the soil due to heavy rainfall and irrigation. The saturated soil conditions, as mentioned by Khayatnezhad and Gholamin[61], can be susceptible to the reduction in leaf chlorophyll content in our study. Full irrigation, in addition to frequent rainfall, might have contributed to a decline in leaf chlorophyll content for the control I1. Although there was no significant difference in biochar treatments in 2021 (Fig. 4C and D), significant differences were observed at 70 DAP and 90 DAP in 2022 where higher biochar rates were contributing to higher chlorophyll synthesis in cucumber (Fig. 4D). Our finding aligns with a previous study Singh, et al.[13]that suggested that the hardwood biochar at 13 t/ha increased chlorophyll SPAD reading compared to the control, providing some N and Mg to leaves, and ultimately, enhancing chlorophyll synthesis in sweet corn. The application of biochar soil amendment can increase nutrient supply and uptake by plants by catalyzing biotic and abiotic reactions in the rhizosphere, which help stimulate plant development[62]. The contribution of biochar in supplying essential nutrients for leaf chlorophyll synthesis in various crops has been reported in different studies[63,64]. Also, higher Ksat, which eases the water flow in the soil in biochar-amended plots, retained nutrients through physical and chemical means, avoiding leaching[65], which plants could uptake for different physiological processes like chlorophyll synthesis.

Stomatal conductance and transpiration

In 2021, at 33 DAP, I4 had the least gs among all the irrigation levels, while the difference in gs remained insignificant among differential irrigations treatments at 62 and 74 DAP (Fig. 5A). In 2022, at 42 DAP, I1 and I2 had comparable gs, which were significantly higher than I3 and I4 (Fig. 5B). At 70 DAP, I3 had significantly

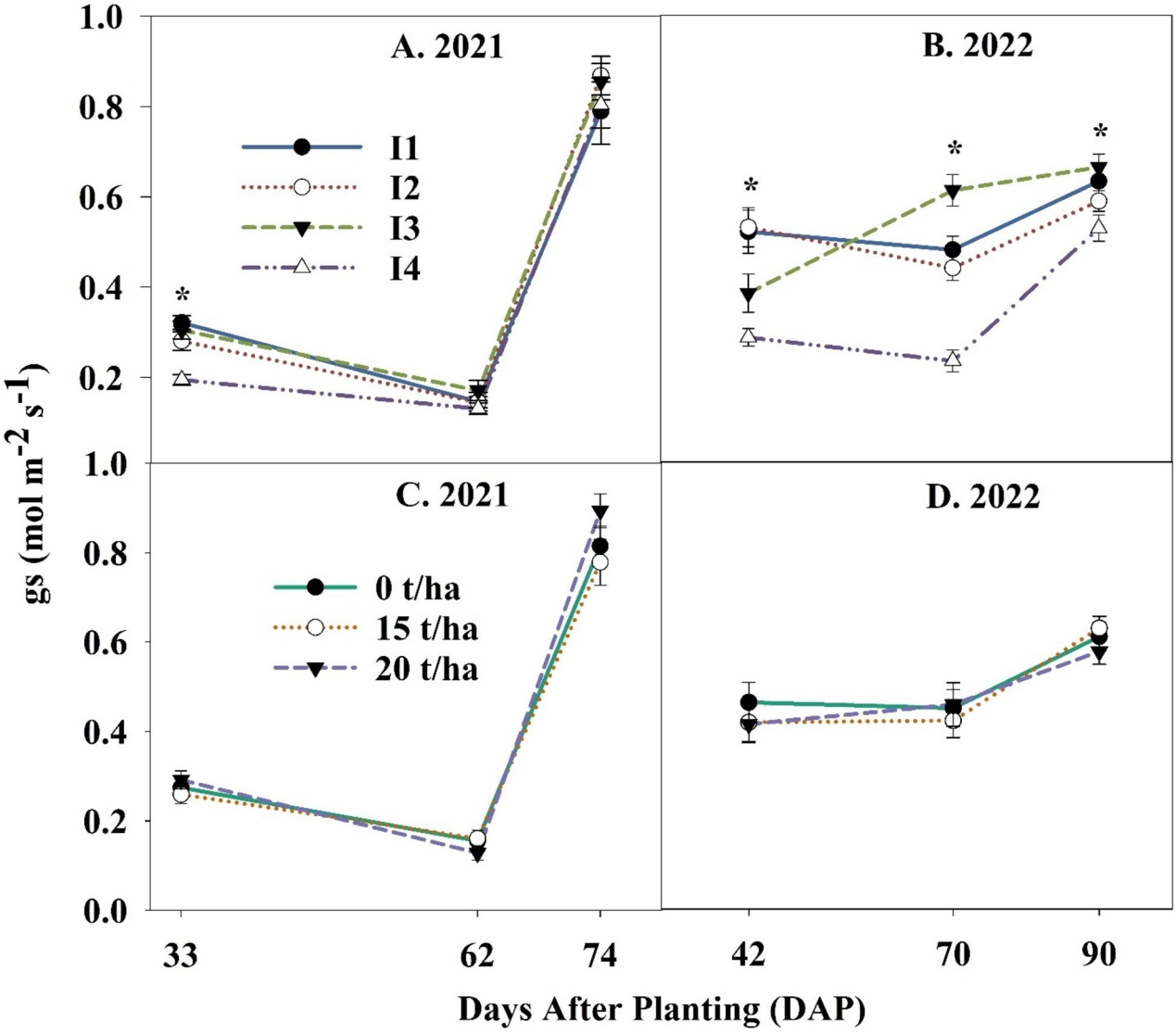

Fig. 5. Effect of deficit irrigation and biochar rates on stomatal conductance (gs) of cucumber in 2021 (**A** & **C**) and 2022 (**B** & **D**) in Lubbock, TX. The bars indicate standard errors and symbol * indicates a significant difference among treatments at $P \leq 0.05$ level of significance.

higher gs than all three irrigation levels, and at 90 DAP, I1 and I3 had comparable gs, followed by I2, and the least was in I4. Notably, the water stress at 80% ETc at any growth stage of cucumber could maintain similar or even higher gs than full irrigation (100% ETc). Therefore, agreeing with previous reports that highlighted the restriction in the opening of stomata under declining water conditions[4,66], our finding also suggests that the cucumber plants likely adapt to water stress or compensate for soil water stress in the root zone imposed by DI level at 80% ETc (i.e., I2 treatment) to maintain similar or higher gs than the control I1 throughout the growing season. Nonetheless, when the irrigation level reached 60% ETc or below, the stomatal conductance in cucumber was lowered, for instance, once the irrigation dropped to 60% ETc at 70 and 90 DAP in 2022 the gs was decreased. A reverse pattern was observed for I3 treatment, where gs was low at the beginning (i.e., 42 DAP)) and increased later (i.e., 70 and 90 DAP) when the irrigation level was maintained at 80% ETc. Biochar treatments did not influence gs responses at different growth stages of cucumber in 2021 and 2022 (Fig. 5C and D).

As shown in Fig. 6A and B, the stomatal closure also reduced transpiration rate (E) of the cucumber plant at 33 DAP in 2021, and 42, 70, and 90 DAP in 2022. The E was significantly lower for I4 treatment at 33 DAP but remained unaffected among DI treatments at 62 and 74 DAP in 2021 (Fig. 6A). A distinct pattern of E to be pointed out in 2022 (Fig. 6B) is that I1 and I2 treatments exhibited comparable E at 42 DAP, and when the second differential irrigation level at 80% ETc was employed to I3, E increased but with 60% ETc, E decreased in I2. Hence, it is envisaged that the transpirational water loss was primarily regulated by stomatal conductance which is dependent on the water availability in leaves and the plant system resulting from different DI levels[4]. The transpiration loss was more distinct among irrigation levels in 2022 compared to 2021 due to the relatively hotter and drier period in 2022. Biochar treatments did not affect transpiration as there was no difference in stomatal conductance under different biochar application rates (Fig. 6C and D).

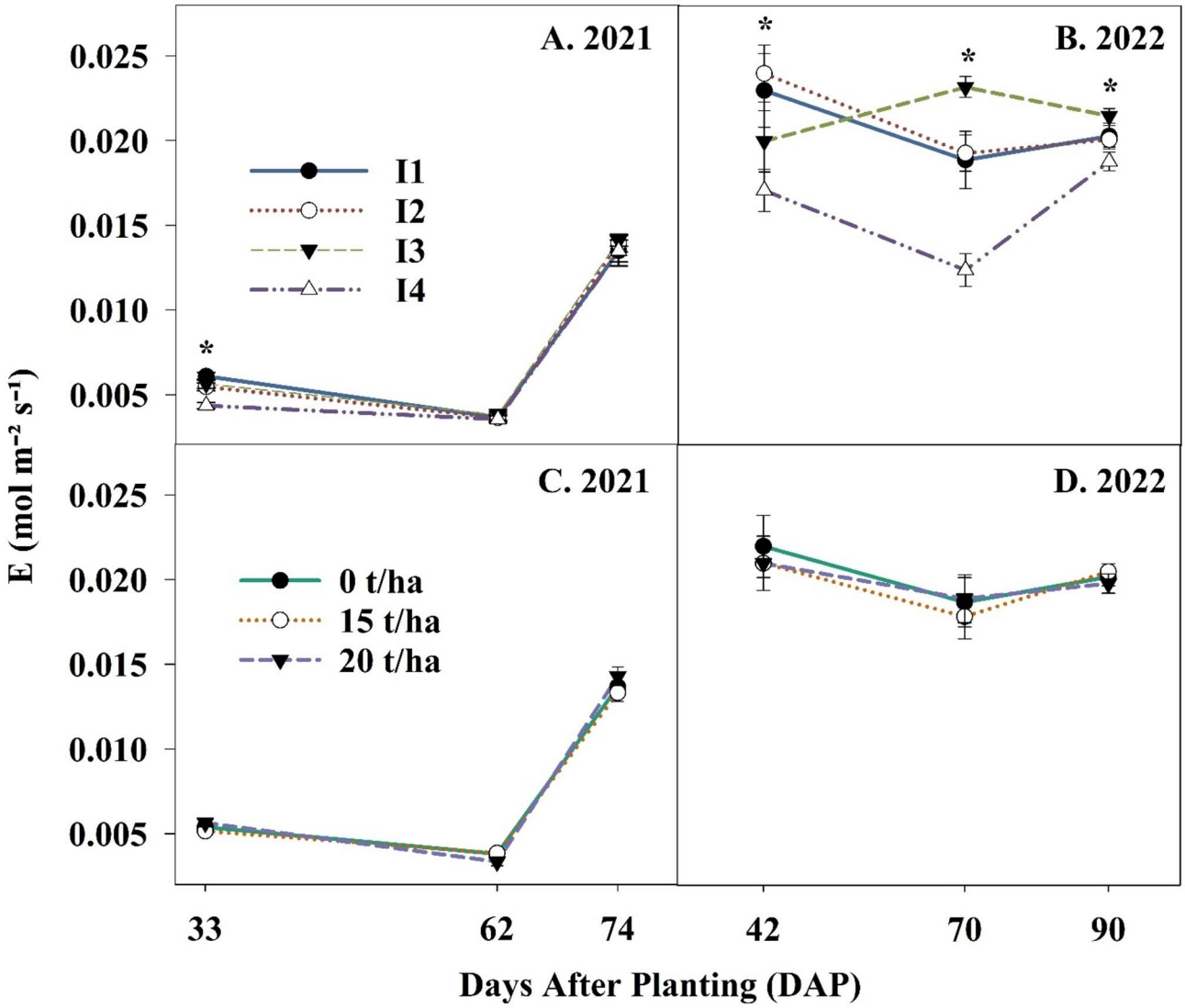

Fig. 6. Effect of deficit irrigation and biochar rates on leaf transpiration (**E**) of cucumber in 2021 (**A** & **C**) and 2022 (**B** & **D**) in Lubbock, TX. The bars indicate standard errors and symbol * indicates a significant difference among treatments at $P \leq 0.05$ level of significance.

Photosynthesis and instantaneous water use efficiency

In 2021, there was comparable net photosynthesis (Pn) between I1 and I3, followed by significantly lower Pn in I2 and the least in I4 at 33 DAP. Still, Pn remained comparable among DI treatments at 62 and 74 DAP (Fig. 7A). In 2022, irrigation levels I1, I2, and I3 had comparable Pn, significantly higher than I4 at 42 DAP (Fig. 7B). At 70 DAP, I3 had the highest Pn followed by I1, and Pn values under I3 and I1 treatments were significantly higher than I2 and I4. The reduced stomatal conductance can be a limiting factor for reduced photosynthetic CO_2 assimilation as the intake is constrained by lower water loss in I2 and I4 treatments[67,68]. At 90 DAP, again, there was no significant difference in Pn among irrigation levels. It is worth noting that I2 treatment for cucumber plants responded favorably by sustaining Pn, i.e., maintaining satisfactory CO_2assimilation for most of the growing period under semi-arid climatic conditions. The gs appeared more sensitive than Pn because gs values were significantly affected under water stress conditions, likely leaving Pn unaffected among different DI levels at 90 DAP in 2022. Our finding agreed with the previous studies that the depression in Pn is entirely triggered with the closure of the stomatal aperture[69,70]. The biochar treatments did not significantly affect Pn in cucumbers in both years (Fig. 7C and D).

Instantaneous WUE (iWUE) was significantly higher in I4 than other irrigation treatments (i.e., I1, I2, and I3) at 33 DAP in 2021, and 42 DAP and 70 DAP in 2022 (Fig. 8A and B). Notably, the percentage decline in gs exceeded the percentage decline in Pn in response to water stress imposed by DI treatments (Figs. 5A-B and 7A-B). The most plausible explanation is that the water stress under different growth stage-based DI treatments largely affects gs ($R^2 > 0.9$) more than Pn ($R^2 < 0.5$), which, as also reported in previous studies[71], later triggers

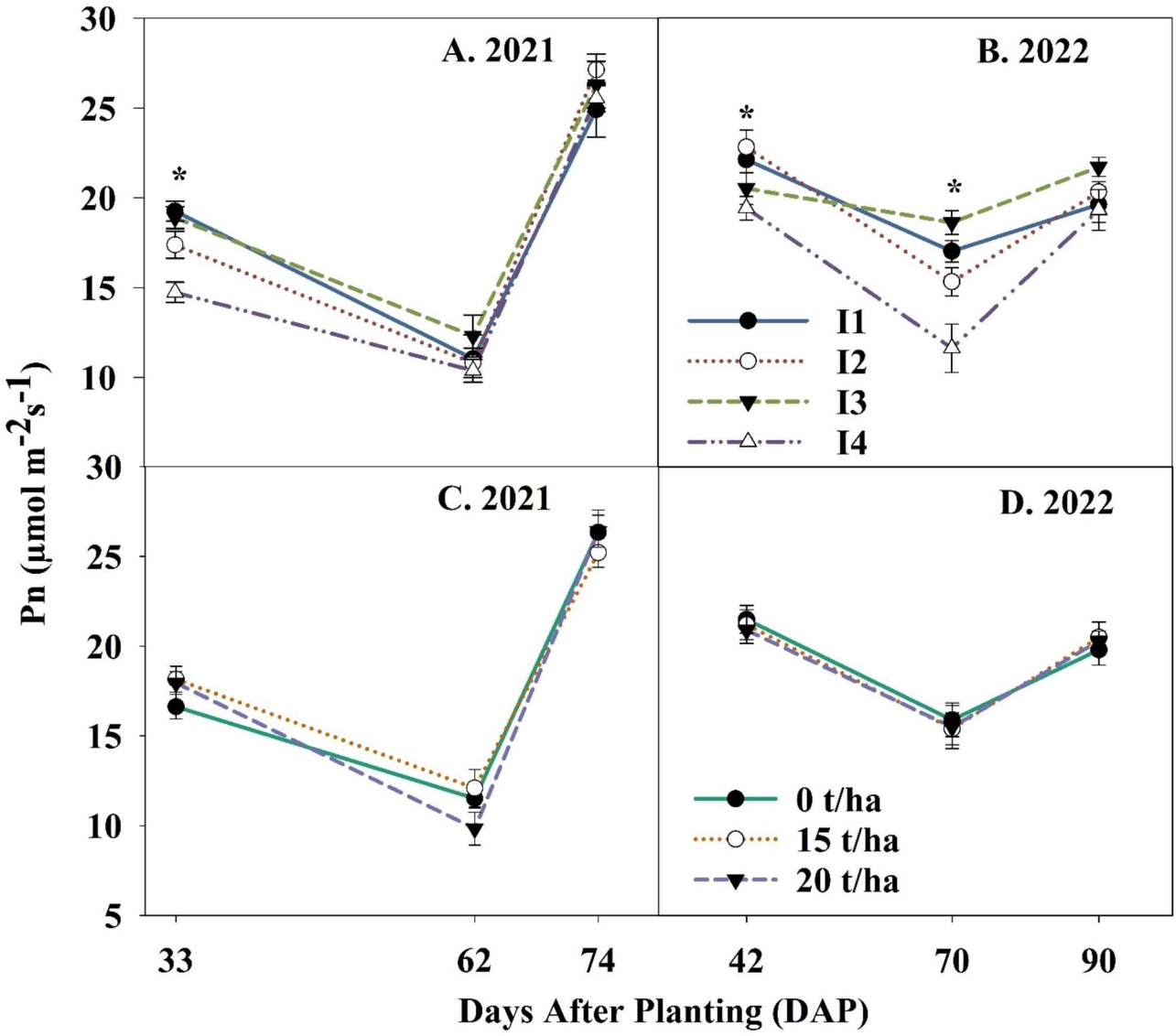

Fig. 7. Effect of deficit irrigation and biochar rates on net photosynthesis (Pn) of cucumber in 2021 (**A** & **C**) and 2022 (**B** & **D**) in Lubbock, TX. The bars indicate standard errors and symbol * indicates a significant difference among treatments at $P \leq 0.05$ level of significance.

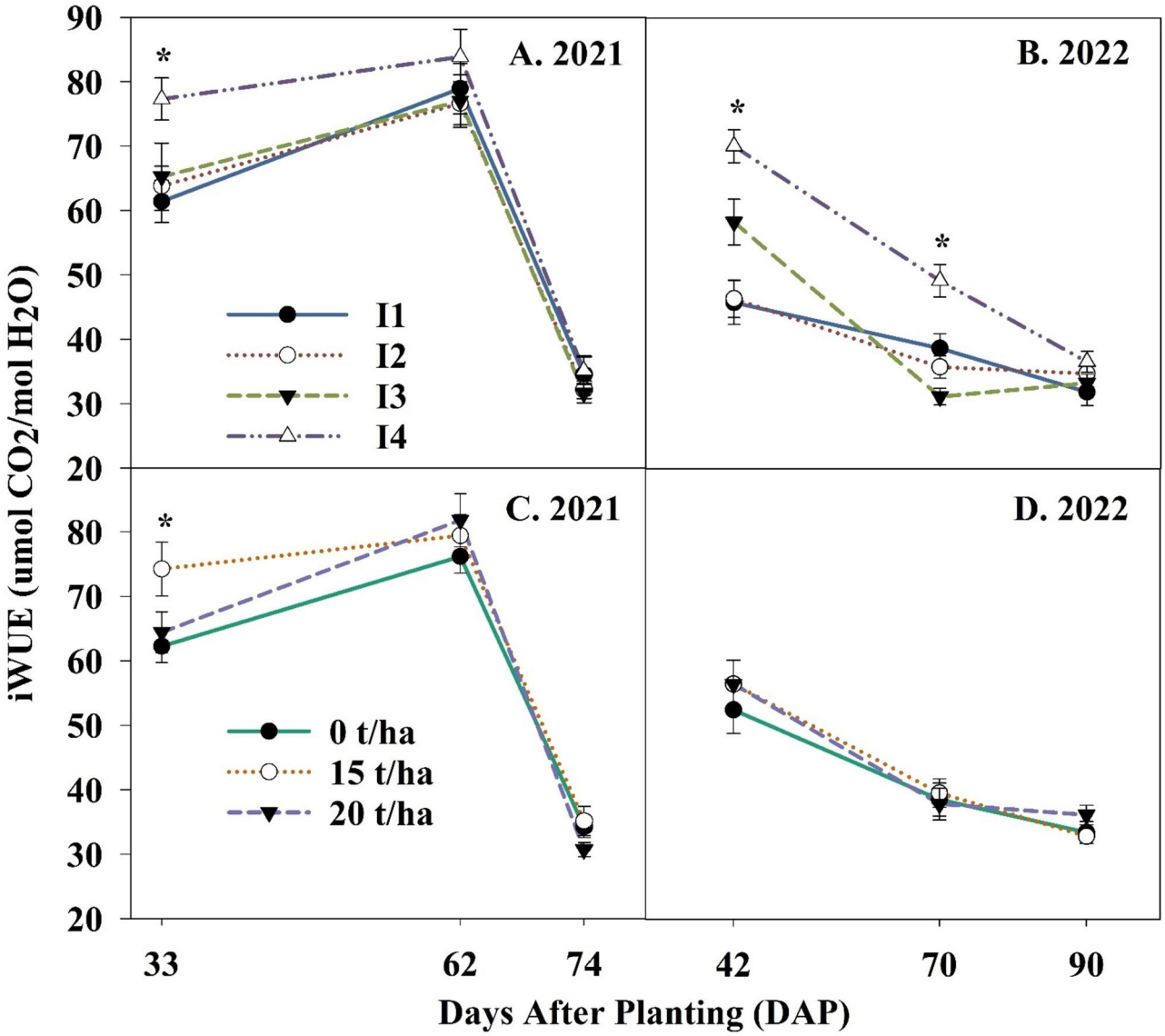

Fig. 8. Effect of deficit irrigation and biochar rates on instantaneous water use efficiency (iWUE) of cucumber in 2021 (**A** & **C**) and 2022 (**B** & **D**) in Lubbock, TX. The bars indicate standard errors and symbol * indicates a significant difference among treatments at $P \leq 0.05$ level of significance.

a reduction in Pn. iWUE increased with increasing water stress levels in different DI treatments, similar to the findings of Singh, et al.[13], indicating that cucumber plants under extreme (I4) water stress conditions could exhibit high Pn activity for a given value of gs, resulting in high iWUE. The biochar application rate at 15 t/ha significantly increased iWUE at 33 DAP only in 2021 (Fig. 8C), while it remained unaffected later in the 2021 growing season and throughout the 2022 growing season (Fig. 8D).

Leaf area index, biomass, and yield

Leaf area index significantly differed among irrigation levels throughout the growing season in both years (Fig. 9A and B). LAI started to increase at 33 DAP, reached its peak at 62 DAP, and decreased later except for I3, which continued to increase until 74 DAP in 2021 (Fig. 9A). The DI level I2 had significantly higher LAI at 33 DAP compared to other DI levels, which was comparable to the control I1 at 62 DAP but decreased at 74 DAP. There was a sharp reduction in LAI in I4 beyond 62 DAP in 2021. In 2022, although the highest LAI was observed for I2 at the beginning, I1 had the highest LAI at 70 and 90 DAP (Fig. 9B). The LAI in all the irrigation levels decreased after 70 DAP. The sharp decline in LAI at the later growth stage of the cucumber plant, except I3 in 2021, could be attributed to senescence effects[72] in both years. As visualized in Fig. 9A and B, the onset of leaf senescence was relatively rapid for the DI treatment that imposed the most water stress conditions, i.e., I4. Our study showed that I2 at the beginning and I3 at the later growth stage when the irrigation levels were at 80% ETc helped to maintain comparable or even higher LAI than full irrigation (i.e., I1). Similar results were obtained by a previous study in semi-arid West Texas Parkash, et al.[4], in which the cucumber plants exposed to

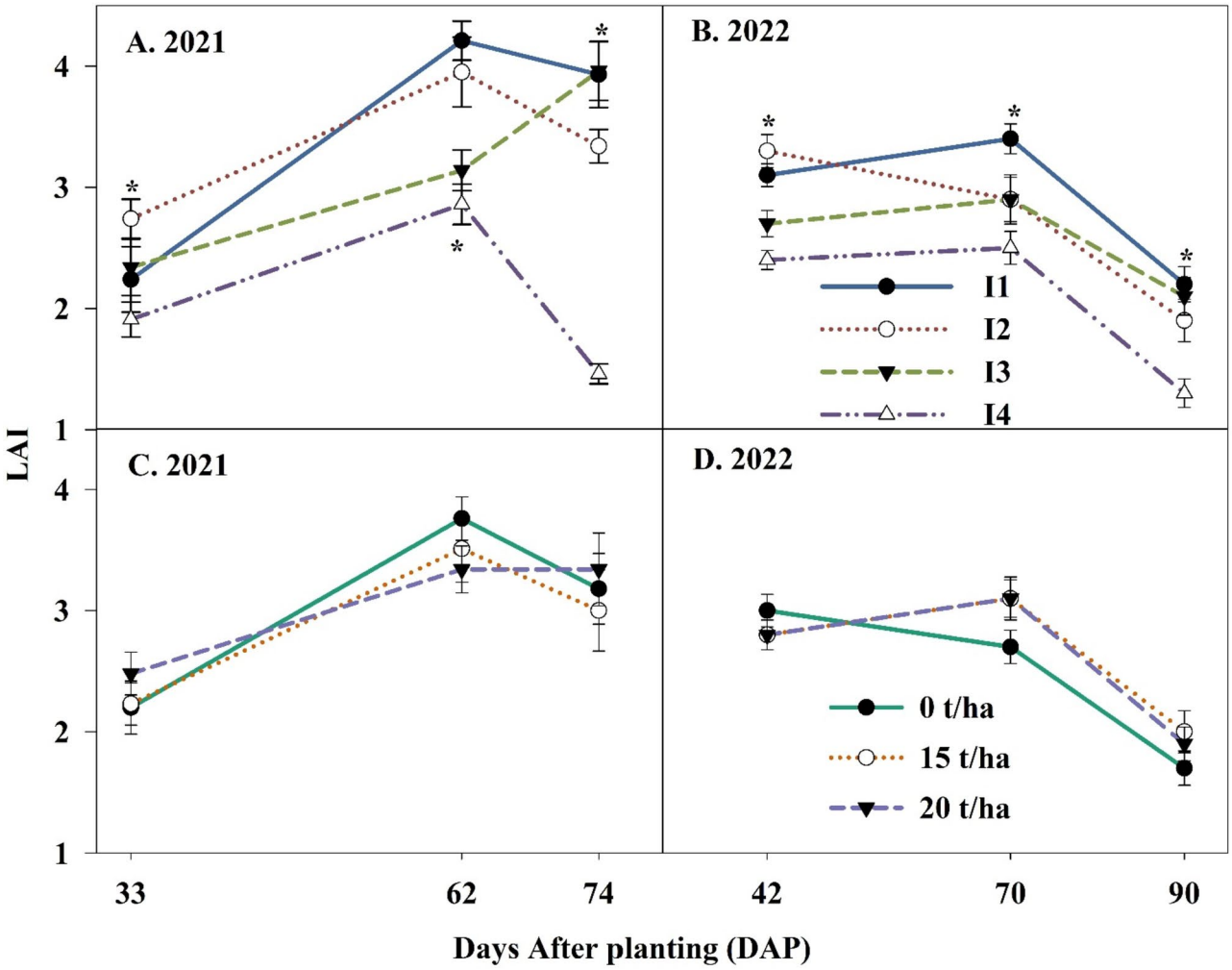

Fig. 9. Effect of deficit irrigation and biochar rates on leaf area index (LAI) of cucumber in 2021 (**A** & **C**) and 2022 (**B** & **D**) in Lubbock, TX. The bars indicate standard errors and symbol * indicates a significant difference among treatments at $P \leq 0.05$ level of significance.

irrigation level of 80% of ETc had enhanced leaf area similar to the cucumber plants growing under irrigation level of 100% of ETc. In contrast, differential irrigation was applied at different growth stages in this study, the cucumber plants. exposed to mild water stress conditions (80% ETc) until the mid-season followed by 60% ETc afterward, maintained satisfactory leaf expansion. Apparently, the growth-based DI levels (i.e., I2) evaluated in this study could maintain adequate leaf area or canopy structure of the cucumber plant that would help the plant prepare more photosynthates and allocate assimilates to the developing fruits. The effect of biochar on the LAI of cucumber remained insignificant in both years (Fig. 9C and D).

Different DI levels affected the vegetative dry biomass of cucumber in both years (Table 2). In 2021, irrigation treatments I1, I2, and I3 produced comparable aboveground vegetative dry biomass, while it was the lowest in I4. However, in 2022, compared to I1, I2 only had comparable vegetative dry biomass with I1, whereas the biomass was significantly reduced under I3 and I4 treatments. Singh-Sangwan, Abad Farooqi and Singh Sangwan[73]mentioned that plants under water stress conditions mobilize the limited water for survival rather than using it for growth. The dry vegetative biomass under I2 compared to I1 might be explained by LAI, an indicator of plant growth and biomass production[74]. The leaf expansion was similar between I1 and I2 treatments towards the end of the cucumber growing season, which might be the reason why I2 produced similar vegetative dry vegetative biomass compared to I1. Unlike the I2 treatment, the DI level I3, which was subject to more water stress conditions for plants (60% ETc until mid-season), did not maintain biomass accumulation in 2022, as limited water conditions restricted the growth. It can be further explained by the fact that the cucumber plants might experience water stress caused by the DI level of 60% of ETc mainly when the plants were actively growing and requiring enough water for growth before the mid-season. But, switching to DI level of 80% of ETc after the mid-season did not contribute significantly to biomass accumulation because the plants had already assimilated the resources to grow fruits. Compared to I1, the irrigation level I4 significantly reduced vegetative dry biomass by 37% in 2021 and 57% in 2022, indicating that, as also suggested in a previous study[4], the cucumber plants under severe water stress (I4) plants undergo growth adjustments as a mechanism to cope with limited water

Year	Factor	Dry vegetative biomass (kg/ha)	Yield (kg/ha)	Individual fruit weight (g)
2021	Irrigation (I)			
	I1	2483a†	48933a	304a
	I2	2443a	41423b	271b
	I3	2417a	38828b	262b
	I4	1565b	23022c	223c
	Biochar (B)			
	0 t/ha	2286a	36908a	265a
	15 t/ha	2275a	37978a	262a
	20 t/ha	2120a	39268a	267a
	Interaction (I x B)	ns	ns	ns
2022	Irrigation (I)			
	I1	2520a	32618a	288a
	I2	2240ab	28357a	302a
	I3	2090b	28875a	286a
	I4	1086c	18113b	242b
	Biochar (B)			
	0 t/ha	1821a	25854a	274a
	15 t/ha	1851a	27188a	283a
	20 t/ha	2280a	27930a	282a
	Interaction (I x B)	ns	ns	ns

Table 2. Effect of deficit irrigation and Biochar rates on dry vegetative biomass, fruit yield, and individual fruit weight of cucumber in 2021 and 2022 in Lubbock, TX. Note: † Mean values followed by different letters within a column are significantly different according to the LSD test at $p \leq 0.05$. 's' signifies significant whereas 'ns' signifies not significant interaction at $p \leq 0.05$.

resources. As most of the growth and physiology of the cucumber plants were unaffected by biochar treatments, the vegetative biomass also remained unaffected in both years.

The individual fruit weight is an important determinant of the economic yield of cucumber[75]. This signifies that the total yield is the function of the sink strength of each fruit. The individual fruit weight was reduced significantly for I2, I3, and I4 treatments by 11, 13, and 27%, respectively, in 2021, but only by 0.5 and 16% for I3 and I4, respectively, in 2022 (Table 2). Alternatively, fruit weight was increased by 5% in I2 in 2022 (Table 2). Our study is in agreement with the findings by Parkash, et al.[4]where the severe water stress imposed by DI level of 40% of ETc had the largest decrease in average fruit weight of cucumber compared to 100% ETc. It was most likely that, as reported by Abd El-Mageed and Semida[76] for cucumber plants, insufficient soil water resulting from DI levels might cause inadequate water and nutrient absorption by cucumber plants that could promote fruit weight. Biochar amendments did not influence the individual fruit weight of the cucumber.

The water stress under different DI treatments significantly affected the yield of cucumber (Table 2). In 2021, the yield penalties in I2, I3, and I4 were 15, 20 and 53%, respectively, but in 2022, the penalties were reduced to 13, 11 and 44%, respectively compared to I1. The average reduction in cucumber yield for the 2021–2022 seasons was 14% for I2, 16% for I3, and 49% for I4 compared to I1. It is noteworthy that the cucumber plants might be sensitive to water stress resulting from the reduced water availability under DI treatment, which could cause yield losses. Nevertheless, the application of DI treatment I2 in the cucumber production system could lead to satisfactory levels of biomass production with less water, i.e., more savings in irrigation water allocation than other DI treatments. Previous studies have also reported yield reductions in cucumbers under DI strategies. For example, Abd El-Mageed, et al.[77] showed that DI at 80% ETc replacement did not significantly reduce the yield but a severe decline occurred in 60% ETc water replacement treatments compared to 100% ETc in field-grown cucumber. Another important aspect to be noted is that the yield reductions under growth stage-based DI levels could vary depending on the growth stage at which the ETc deficit was imposed. For example, for the DI treatment I2, 80% ETc until the mid-season followed by 60% after the mid-season had caused a lower, i.e., a 15% yield reduction compared to 60% ETc until mid-season followed by 80% ETc, accounting for a 20% yield reduction in 2021. Mao, et al.[17]identified the mid-stage of cucumber to be more sensitive than the later stage for water stress impacting yield. Water stress could lead to flower abortion in cucumbers[22]. When there is higher water stress in cucumbers, the flowers tend to abort, leading to lower yields. As noticed during the routine measurements of plant physiological parameters, until the mid-season, the growth was rapid allowing a higher number of fertile flowers at the nodes. Once it crossed the mid-season, the growth was limited, with fewer nodes producing new flowers.

The biochar treatments 15 t/ha and 20 t/ha increased the yield by 3 and 6%, respectively, in 2021 and by 5% and 8%, respectively, in 2022 compared to the control (0 t/ha). However, the increase in yield was not statistically significant. This minor yield increment could be partially attributed to the enhanced saturated hydraulic conductivity resulting from added biochar amendment, which might regulate infiltration and improve water

movement and redistribution of soil water in the root zone, likely influencing adequate root water and nutrient uptakes[78].

Conclusions

This study investigated the potential effects of integrating growth stage-based deficit irrigation (DI) levels and biochar application rates on cucumber productivity in semi-arid West Texas. There were no significant interactive effects between irrigation and biochar application on cucumber performance. However, with increased water stress conditions for the plants imposed by comparatively lower DI level treatments at various growth stages (between establishment and mid-season and mid-season and maturity), there was a decline in chlorophyll content, stomatal conductance, transpiration, and net photosynthesis in cucumber. Leaf area index (LAI) and vegetative dry biomass were comparable between I1 and I2, suggesting the adaptability of cucumber plants under DI-imposed water stress conditions by applying irrigation level of 80% of crop evapotranspiration (ETc) until the mid-season followed by irrigation level of 60% of ETc after the mid-season. Yield reductions in cucumbers were only 15 and 20% in I2 and I3, respectively, in 2021, but it narrowed down to 13 and 11% in 2022 compared to I1. The biomass production for DI treatment I2 was similar to I1 (i.e., full irrigation or control treatment), with a minor reduction in cucumber yield and the potential for saving approximately 25% of irrigation water without compromising plant growth and physiological parameters. Therefore, DI levels that maintain moderate water stress conditions for the cucumber plants, especially I2, can be adopted as a promising growth stage-based DI strategy in semi-arid environments such as West Texas. The application of biochar amendment at higher rates (15 and 20 t/ha) significantly increased the saturated hydraulic conductivity for semi-arid soil with relatively higher clay contents in the second growing season. The application of biochar, integrated with growth stage-based DI, showed limited benefits to cucumber growth, physiology, and yield in both growing seasons. Our study provides unique information on managing irrigation for sustainable production of cucumber in water limited regions like West Texas which will help vegetable growers, water users and policy makers to manage water resources. This study also creates an opportunity for soil management using biochar under limited irrigation for vegetable production. However, studies aimed at assessing long-term responses of cucumber are suggested to quantify the potential of integrating biochar and DI strategies for sustaining cucumber production in semi-arid environments.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Conceptualization: A.K., S.S.; Methodology: A.K.; Data collection: A.K., M.S., P.B.; Data Curation: A.K.; Software and Analysis: A.K.; Writing -Original draft preparation: A.K.; Supervision: S.S.; Review and editing: S.S., M.S., S.D., C.S., G.R. All authors reviewed the manuscript.

Declarations

Competing interests
The authors declare no competing interests.

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